

Scholastic Aptitude Test, 2015 EC, natural science, entrance exam, round 1

Grade 12 · Aptitude · 2015 E.C.

Time allowed: 120 minutes

Questions: 42

Total marks: 42

Name _____ Grade _____ ID No. _____ Date _____

1. Pessimism (1)
 - A. Sorrow
 - B. Optimism
 - C. Arrogance
 - D. Despair
2. Polygamy (1)
 - A. Monogamy
 - B. Interference
 - C. Recovery
 - D. Detour
3. Prospect (1)
 - A. Retrospect
 - B. Traditional
 - C. Ideal
 - D. Modernist
4. The diffusion of the virus had reached an epidemic proportion. (1)
 - A. dissemination
 - B. spread
 - C. diagnosis
 - D. vulnerability
5. The electric poles were impregnated with chemical solutions for durability. (1)
 - A. saturated
 - B. varnished
 - C. erected
 - D. washed
6. The sense of animosity between the rival groups was resolved peacefully. (1)
 - A. grievance
 - B. hatred
 - C. exposure
 - D. revenge
7. The practice of tax evasion is harmful to the development of a nation. (1)
 - A. avoidance
 - B. endorsement
 - C. conviction
 - D. resilience
8. GUNPOWDER: EXPLOSION; Volcano: _____ (1)
 - A. Landslide
 - B. Concentration
 - C. Eruption
 - D. Pasteurization
9. CARAVAN: DESERT; Perch: _____ (1)
 - A. Aquatic
 - B. Terrestrial
 - C. Aerial
 - D. Tropical
10. HISTORY: CHRONOLOGICAL; Topography: _____ (1)
 - A. Temperature
 - B. Spatial
 - C. Climate
 - D. Travel

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11. DREAM: SHATTERED; Promises: _____ (1)
- | | |
|--------------|--------------|
| A. Broken | B. Denied |
| C. Cancelled | D. fulfilled |
-
12. BROADCAST : VERBAL; Paralanguage: _____ (1)
- | | |
|-----------------|---------------|
| A. Audio-visual | B. Imagery |
| C. Body | D. Multimodal |
-
13. The spread of English across the globe is attributed to _____ (1)
- | | |
|--------------------------------|------------------------|
| A. WWII | B. British colonialism |
| C. the Will of Queen Elizabeth | D. WWI |
-
14. What was the misconception of Indian Head Masters about the end goal of education? (1)
- A. Linguistic hybridity
- B. Mastering the 'Queen's English'
- C. Reading Shakespeare
- D. A preference for the American English
-
15. The word recedes in paragraph 2, could interchangeably be used with _____ (1)
- | | |
|-----------------|---------------|
| A. outshines | B. shrinks |
| C. predominates | D. outnumbers |
-
16. The word monolithic in paragraph 2 is synonymous with _____ (1)
- | | |
|------------|-----------------|
| A. hybrid | B. multilingual |
| C. unitary | D. pure |
-
17. Statement: In many undemocratic countries, minority groups were marginalized for one reason or another. (1)
- Conclusions: I. In such countries, individual rights were more important than group rights. II. In such countries, group rights were more important than individual rights. III. In such countries, citizens could be marginalized on racial and religious grounds. IV. Collective identity was more significant than individual identity. Which one of the conclusions is more logical?
- | | |
|--------------------------|------------------------------|
| A. Conclusion I | B. Conclusion II |
| C. Conclusions I and III | D. Conclusion II, III and IV |
-
18. Statement: Many investment projects contribute to environmental pollution unless they are closely monitored by regulatory agencies. Conclusion: I. The development policies of certain countries is exemplary. II. Environmental impact should not be undermined in the implementation of projects. III. Regulatory agencies are responsible for monitoring the implementation of modalities. Which one of the conclusions is implied by the head sentence? (1)
- | | |
|---------------------------|------------------------|
| A. Conclusion I | B. Conclusion II |
| C. Conclusions II and III | D. Conclusion I and II |
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19. The migration of birds from one part of the world to another is attributed to impulses of weather. They flock from Polar Regions to the tropical zone or vice-versa following seasonal changes. The choice of their destination depends on whether or not these migratory birds are cold-blooded. The assumption of the above argument revolves around _____ (1)
- A. the seasonal readjustment of cold-blooded birds
- B. the bipolar migration of all birds
- C. the survival of birds under the strains of climate
- D. the difference between polar and tropical birds

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- 20.** The traditional view about the predestination of poverty has been challenged on scientific grounds. With the advent of science and technology, the stratification of the human society into the poor and the rich is based on man-made problems rather than Divine intervention. In order to ensure equality and equitability of natural resources, modern governments should adopt and enforce the pillars of good governance. This writer proposes that (1)
- A. the fate of the poor is irreversible
B. the fortune and the misfortune of humanity is predestined
C. poverty is man-made and can be resolved by humans themselves
D. the traditional view of pessimism is still persistent
-
- 21.** Statement: All public and private schools were closed down during the outburst of the Covid-19 pandemic, which also led to socio-economic crisis. Conclusions: I. The spread of Covid-19 was caused by the closure of schools. II. The spread of Covid-19 had forced the closure of schools apart from impacting the socio-economic sector. III. The outbreak of this virus had nothing to do with the social service sector. The writer seems to underline the consequences of Covid-19 as stated in (1)
- A. conclusion I
B. conclusion II
C. conclusion III
D. Not given
-
- 22.** The Ministry of Inland Revenue and Customs praised the loyal tax payers for discharging their duty. What can we deduce from this premise? (1)
- A. Tax avoidance has completely been put under control
B. Income generation and business are not interrelated
C. There are duty-bound tax payers who support the development endeavours
D. Deduction of value added taxes on the price of goods is fashionable
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- 23.** Statement: The outbreak of malaria in the tropical zones has posed a threat to public health and the GDP of malarial countries. Conclusions: I. Tropical countries should prioritize the eradication of malaria for humanitarian and economic reasons. II. The death of people from malaria does not affect the GDP of a country. III. Malaria should not be the focal point of research. Which one of the above conclusions could be drawn from the premise? (1)
- A. Conclusion II
B. Conclusion I
C. Conclusion III
D. Not given
-
- 24.** Which one of the following is true for every natural number n ? (1)
- A. $(n + 1)^3 \geq n^3 + 1$
B. $6^n - 1 \leq 625$
C. $(n + 1)^2 \leq n^2 - 2n$
D. $n^2 \geq n!$
-
- 25.** If your total income is 12,000 Birr per month and you can't spend more than 60% on housing, what is the maximum amount of your total income that can be spent on housing? (1)
- A. 9,000 Birr
B. 5,600 Birr
C. 6,000 Birr
D. 7,200 Birr
-
- 26.** Suppose that the perimeter of a rectangular farm is 80 meter. If the farm is y meter long and x meter wide then which one of the following functions of x will model the total area of the farm in square meter? (1)
- A. $f(x) = -x^2 + 40x$
B. $g(x) = x^2 - 40x$
C. $h(x) = -2x^2 + 80x$
D. $k(x) = 2x^2 + 80x$

27. Which one of the following is the coefficient of x^3 in the expansion of the algebraic expression $(x + y)^4$? (1)
- A. 2
B. 6
C. 4
D. 3

28. Which one of the following is the degree of the polynomial function $f(x) = (2x + 7)^5(3x - 1)^2$? (1)
- A. 6
B. 5
C. 7
D. 2

29. Which one of the following is necessarily true about any real numbers x and y ? (1)
- A. $(x - y)^2 \leq (x + y)^2$
B. $(x + y)^2 \leq (x - y)^2$
C. $|x + y| \geq |x - y|$
D. $|x + y| = |-x - y|$

30. Suppose $\{a_n\}_{n=1}^{\infty}$ is an arithmetic sequence and $a_5 + a_8 = 41$ and $a_8 - a_5 = 9$, then which one of the following is the sequence? (1)
- A. 2, 5, 8, 11, 14, ...
B. 4, 7, 10, 13, 16, ...
C. 1, 4, 7, 10, 13, ...
D. 3, 6, 9, 12, 15, ...

31. If the n th-term of the arithmetic sequence $\{a_n\}_{n=1}^{\infty}$ is given by $a_n = 3n - 2$, then which one of the following is the sum of the first seven terms of the sequence? (1)
- A. 70
B. 19
C. 72
D. 28

32. Which one of the following is the sum of the first six terms of the geometric sequence (1)

$$\left\{ 3 \left(\frac{1}{2} \right)^{n-1} \right\}_{n=1}^{\infty} ?$$

- A. $\frac{93}{16}$
B. $\frac{189}{32}$
C. $\frac{45}{8}$
D. $\frac{381}{64}$

33. Consider the system of linear equations $x - 2y = 3$ and $3x + 2y = 1$. Then which one of the following is true about the system? (1)
- A. The system has no solution.
B. The system has infinite solutions.
C. The system has two different solutions.
D. The system has unique solution.

34. Suppose you want to select a 3-person committee from a group of 7 teachers. How many ways are there to do this? (1)
- A. 210
B. 35
C. 60
D. 120

35. A box contains 12 pieces of paper, each of which has the name of one of the 12 students in a class consisting of 7 male and 5 female students, all with different names. What is the probability of a randomly selected piece of paper has a female student name? (1)
- A. $\frac{5}{12}$ B. $\frac{5}{7}$
C. $\frac{7}{12}$ D. $\frac{5}{10}$
-
36. According to a survey, in an area, 25% of the trees planted did not grow in the year 2011 E.C. and 35% of the trees planted did not grow in the year 2014 E.C. What was the percentage increase in the fraction of trees planted that did not grow from the year 2011 E.C. to the year 2014 E.C.? (1)
- A. 60% B. 40%
C. 50% D. 30%
-
37. A lake was 30m deep. Two months later, its depth decreased by 20%. If the depth of the lake increased from where it was after two months by 15% in three months' time, what was the cumulative percentage change in the depth of the lake during the five months? (1)
- A. 6% B. 5%
C. 8% D. 10%
-
38. Which one of the following is the solution set of the system of linear equations given by $x + y = 1$ and $3x - y = 7$? (1)
- A. $\{(-1, 2)\}$ B. $\{(2, -1)\}$
C. $\{(2, 1)\}$ D. $\{(-2, 1)\}$
-
39. Which one of the following ordered pair of numbers is not in the solution of the system of linear inequalities $x + y \geq 1$ and $x - 2y \leq 2$? (1)
- A. (1, 2) B. (2, 1)
C. (-1, 2) D. (2, -1)
-
40. Given the system of linear inequalities $ax + by \geq c$ and $ax + by \leq c$ for arbitrary constants a, b and c in $\mathbb{R}$ with $c \neq 0$, then which one of the following is true about the solution set of the system? (1)
- A. $\{(x,y): ax + by \geq c\}$ B. $\{(x,y): ax + by \leq c\}$
C. $\{(x,y): ax + by = c\}$ D. $\{(x,y): ax + by \neq c\}$
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41. A woman spent one fifth of her salary for clothing, two fifth her salary for different bills and 100 Birr for her transport. After covering all these costs she is left with 4,500 Birr. What was the salary she received? (1)
- A. 11,500 Birr B. 12,500 Birr
C. 13,500 Birr D. 10,500 Birr
-
42. During the sunshine, an upright building 16 meters tall is casting a shadow of 12 meters long. At the same time, a nearby upright building is casting a shadow of 18 meters long. If the lengths of the shadows are proportional to the heights of the buildings, what is the height, in meter, of the taller building? (1)
- A. 32 B. 17
C. 36 D. 24

